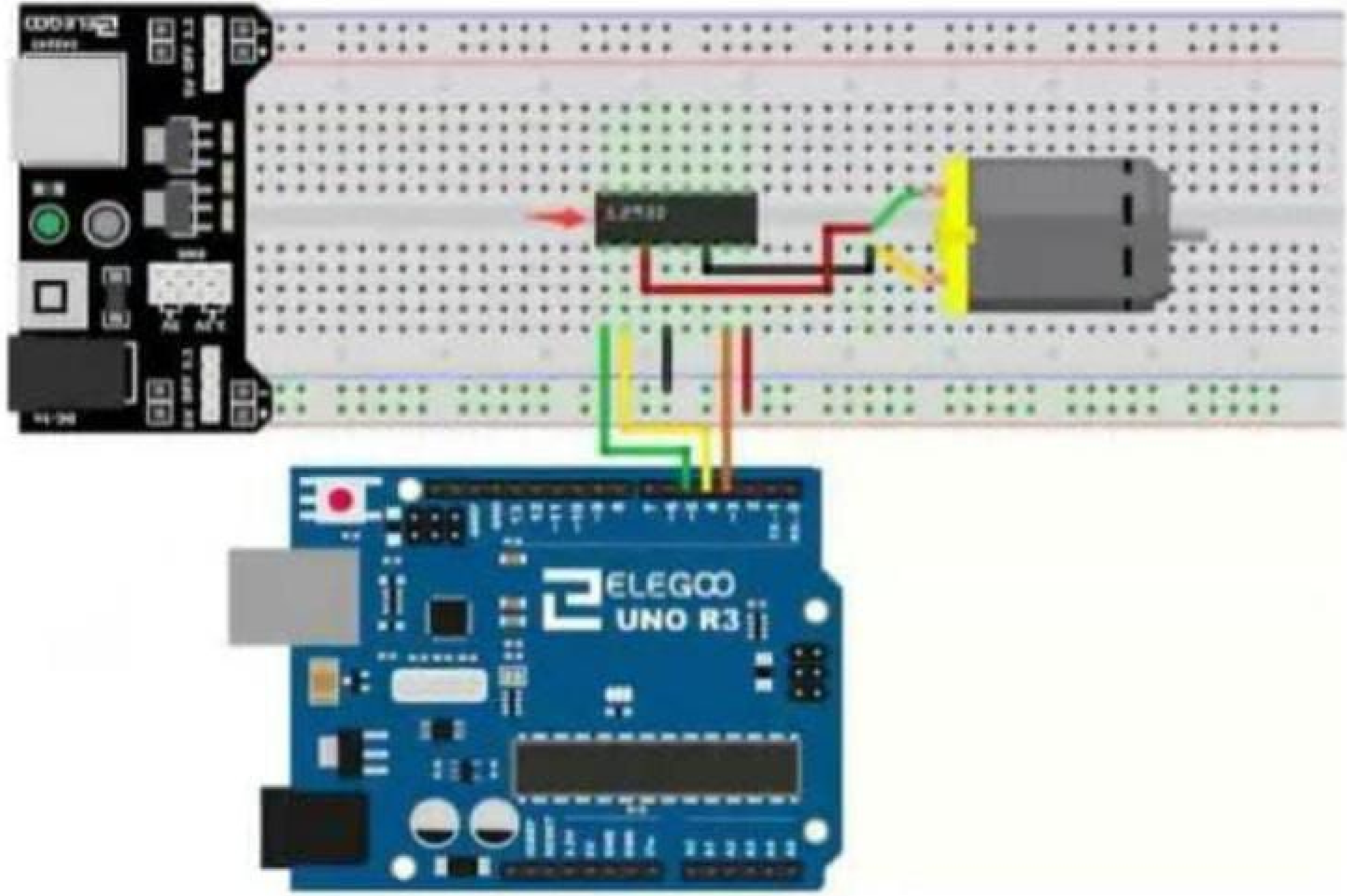
## Board connection




dot.

sketch_aug3b.ino

```
int SpeedPin=5; //EnA (to control the motor speed)
int dir1=4;//In1 (to control the forward direction of
int dir2=3; //In2( to control backward direction of tl
int redled=8;
int greenled=9;

void setup() {
  // put your setup code here, to run once:
pinMode(SpeedPin,OUTPUT);
pinMode(dir1,OUTPUT);
pinMode(dir2,OUTPUT);
pinMode(redled,OUTPUT);
pinMode(greenled,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:

}
```

sketch_aug3b.ino

```
int redled=8;
int greenled=9;

void setup() {
  // put your setup code here, to run once:
pinMode(SpeedPin,OUTPUT);
pinMode(dir1,OUTPUT);
pinMode(dir2,OUTPUT);
pinMode(redled,OUTPUT);
pinMode(greenled,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:

}

void DirectionControl(){ //function to control the pi


}

```

```
void DirectionControl(){ //function to control the spi

analogWrite(SpeedPin,255); //set the motor to the maxi
digitalWrite(dir1,HIGH);
digitalWrite(dir2,LOW);
digitalWrite(greenled,HIGH);
delay (5000);

//now we change the motor's direction
digitalWrite(dir1,LOW);
digitalWrite(dir2,HIGH);
digitalWrite(greenled,LOW);
digitalWrite(redled,HIGH);
delay(5000);

//turn off the motor
digitalWrite(dir1,LOW);
digitalWrite(dir2,LOW);
digitalWrite(dir1,LOW);
digitalWrite(dir1,LOW);

```

sketch_aug5b.ino

```
digitalWrite(dir1,LOW);
digitalWrite(dir2,HIGH);
digitalWrite(greenled,LOW);
digitalWrite(redled,HIGH);
delay(5000);

//turn off the motor
digitalWrite(dir1,LOW);
digitalWrite(dir2,LOW);
digitalWrite(redled,LOW);
}

void SpeedControl (){ //function to control the DC Mo

//Turn on the motor forward
digitalWrite(dir1,HIGH);
digitalWrite(dir2,LOW);

//Accelerate the motor speed in dir  from  0 to 255
for(int i=0; i<=255;i++){
  analogWrite(SpeedPin,i)
}
```

```
void DirectionControl(){ //function to control the spi

analogWrite(SpeedPin,255); //set the motor to the maxi
digitalWrite(dir1,HIGH);
digitalWrite(dir2,LOW);
digitalWrite(greenled,HIGH);
delay (5000);

//now we change the motor's direction
digitalWrite(dir1,LOW);
digitalWrite(dir2,HIGH);
digitalWrite(greenled,LOW);
digitalWrite(redled,HIGH);
delay(5000);

//turn off the motor
digitalWrite(dir1,LOW);
digitalWrite(dir2,LOW);
digitalWrite(redled,LOW);
}
```

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sketch_aug3b.ino

```
//Decelerate the motor speed in dir 1 from 255 to 0
for(int i=255; i>=0;i--){
  analogWrite(SpeedPin,i);
}

//Turn off the motor
digitalWrite(dir1,LOW);
digitalWrite(dir2,LOW);

//Turn on the motor baclward
digitalWrite(dir1,LOW);
digitalWrite(dir2,HIGH);

//Accelerate the motor speed in dir 2  from  0 to 255
for(int i=0; i<=255;i++){
  analogWrite(SpeedPin,i);
}
//Decelerate the motor speed in dir 1 from 255 to 0
for(int i=255; i>=0;i--){
  analogWrite(SpeedPin,i);
}
}
```

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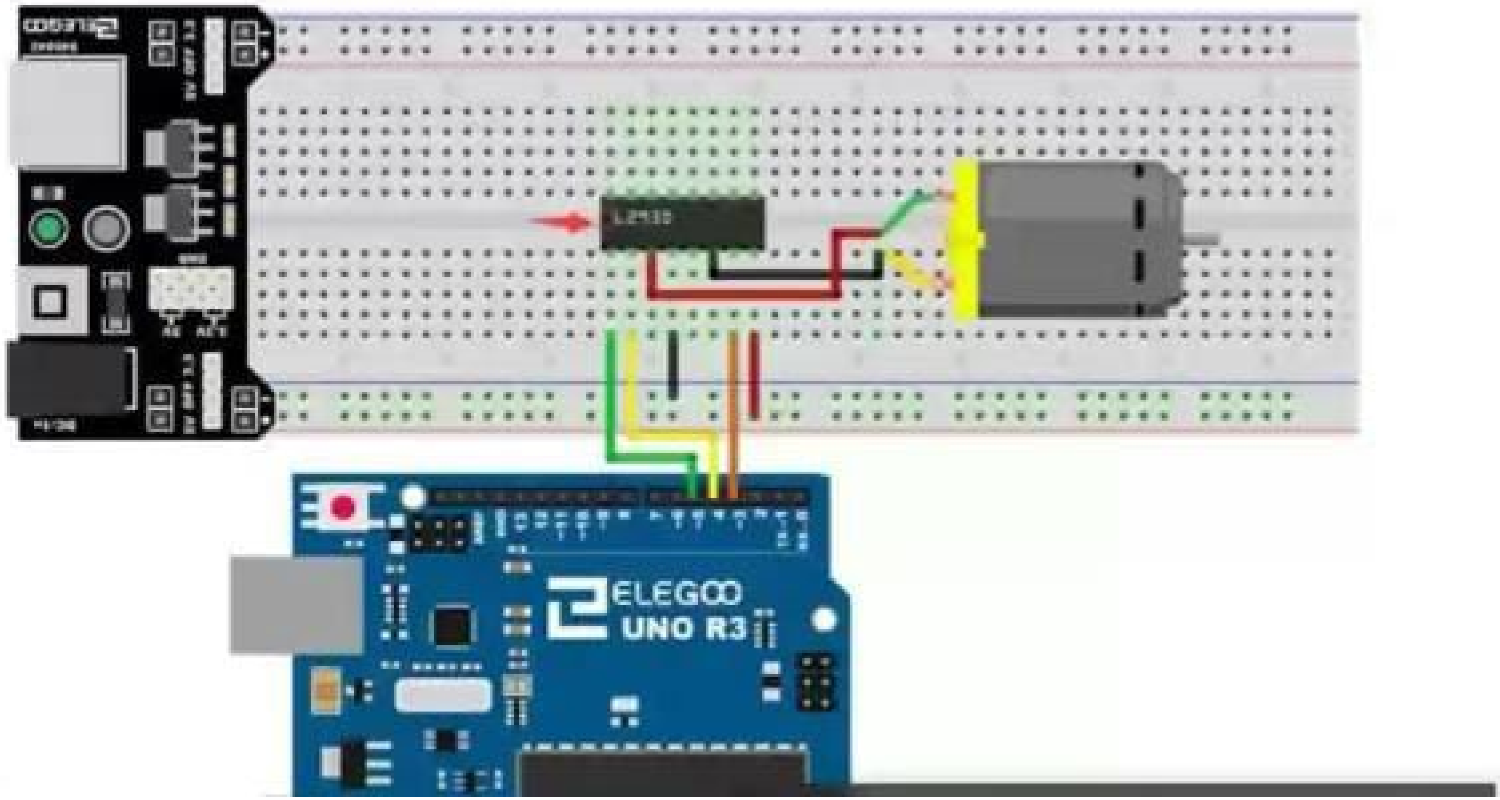
# Board connection



- Click to add text

Define Sensors & Types of sensors

Use the Photo resistor with Arduino

DHT Sensor

- Click to add text

- Click to add text

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# Sensors

Sensors are a device which detects or measures a physical property and records, indicates, or otherwise responds to it. The sensor can have different forms or accessories depending on the surrounding conditions that it should test.


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## Sensors Categories

Sensors can be divided into three main categories:

1. 3-pin Analog Sensor
2. 3-pin Digital Sensor
3. Analog/Digital Sensor

## Sensors Categories

These categories are divided according to the type of the sensor. Does it send a digital signal? Or an Analog signal? Or it uses both analog and digital signals?

We can say that a sensor with three legs, is either an analog or a digital sensor. But a sensor with four legs is an Analog/Digital sensor. In order to differentiate between 3-pin analog and 3-pin digital sensor, we can read the pin labeling.

## Digital Sensor

A digital sensor is a type of sensor that produces a discrete output signal, typically in the form of binary values (0s and 1s) . These sensors convert physical quantities into digital signals that can be easily processed by microcontrollers and digital systems.

### Sensors character

Digital sensors provide output in distinct levels, usually representing "on" (high) and "off" (low) states.

### Applications

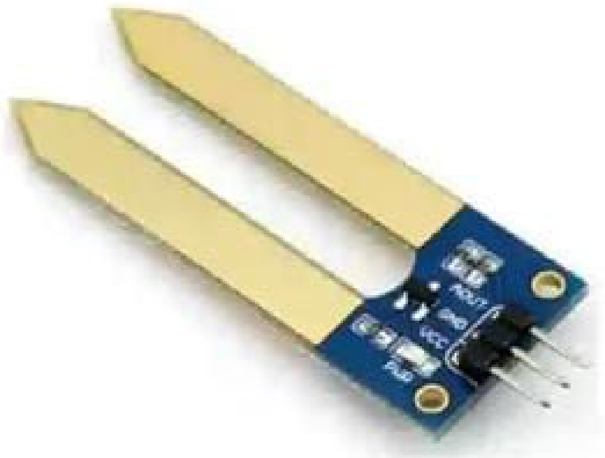
Digital sensors are widely used in various applications, including robotics, IoT devices, and home automation systems.

# Analog Sensor

An analog sensor is a type of sensor that produces a continuous signal or variable output, representing physical quantities such as temperature, light intensity, pressure, or sound.

### Connection method

This sensor should have one pin to the VCC (+5 V) and another to (+0 V). While the third pin should be connected to the analog input.



### Sensors character

Some sensors may have an output range between 0-255 while some can vary between 0-1023.

### Applications

Analog sensors are commonly used in various applications, including weather stations, home automation, and industrial monitoring systems.

# Analog/Digital Sensor Characters

**Digital/Analog sensors have some features that facilitate the execution of tasks.**

1- The analog pin sends a voltage signal that has the actual value. This will give the user a full knowledge about the voltage.

2- The Digital pin sends a signal with either HIGH or LOW Voltage. The signal is considered to be HIGH if the voltage is above the threshold.

3- The sensor has a third option which is the trigger level where the Arduino will only log the values if they are above or below the threshold.

4- The Sensor board is also featured with a power LED and a Status LED, the status LED shows if the digital output is triggered or not.

## Pins Characters of Digital/Analog Sensors

1. AO: Analog Out
2. G: Ground
3. Positive Supply
4. DO: Digital Out

# Activity 1: Photoresistor

## Photoresistor with Arduino

Required Elements:

- Arduino
- X2 Resistors 330 Ω
- Red LED
- Photoresistor
- Breadboard

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## Getting to Know Your Photosensor

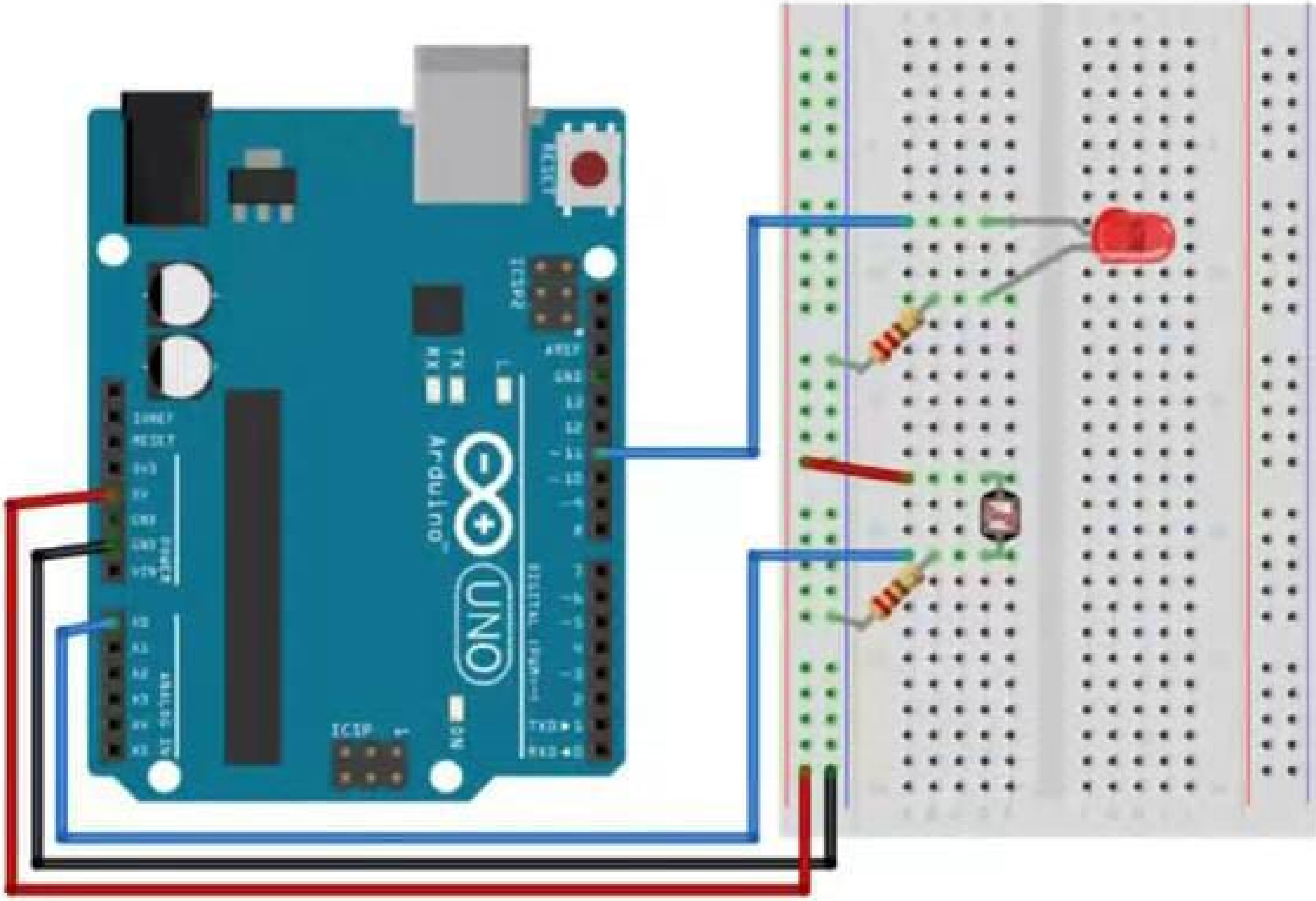
### What is a photo resistor?

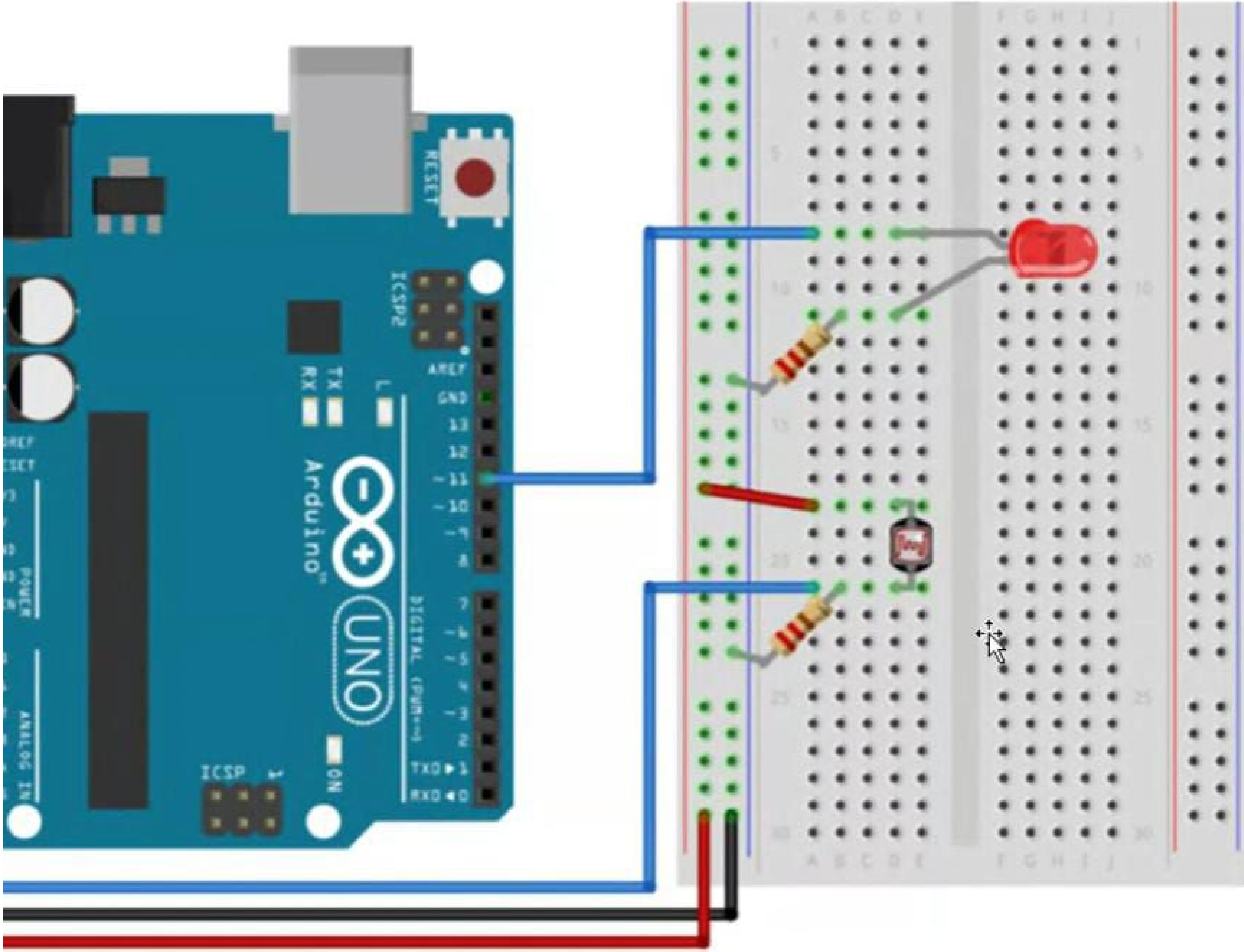
A photo resistor or photocell is **a light-controlled variable resistor**. It is a variable resistor that changes resistivity according to the light intensity. The resistance of a photo resistor decreases with increasing incident light intensity.

### *What are the uses of photo resistors?*

Photo resistors are most often used as **light sensors**. They are often utilized when it is required to detect the presence and absence of light or measure the light intensity. Examples are night lights and photography light meters.

# Board connection




RESET
ICSP2
RX
TX
L
AREF
GND
13
12
~11
~10
Arduino
UNO
DIGITAL (PWM~)
TX0 ▶ 1
RX0 ◀ 0
ICSP
ON
POWER
ANALOG IN

sketch_aug3c.ino

```
int photoresistor=A0;
int led=9;

void setup() {
  // put your setup code here, to run once:
pinMode(led,OUTPUT);
Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
int SensorValue=analogRead(photoresistor);
Serial.print("Sensor Value :");
Serial.println("SensorValue");
delay(1000);
}
```

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dit Sketch Tools Help

Arduino UNO

sketch_aug3d.ino

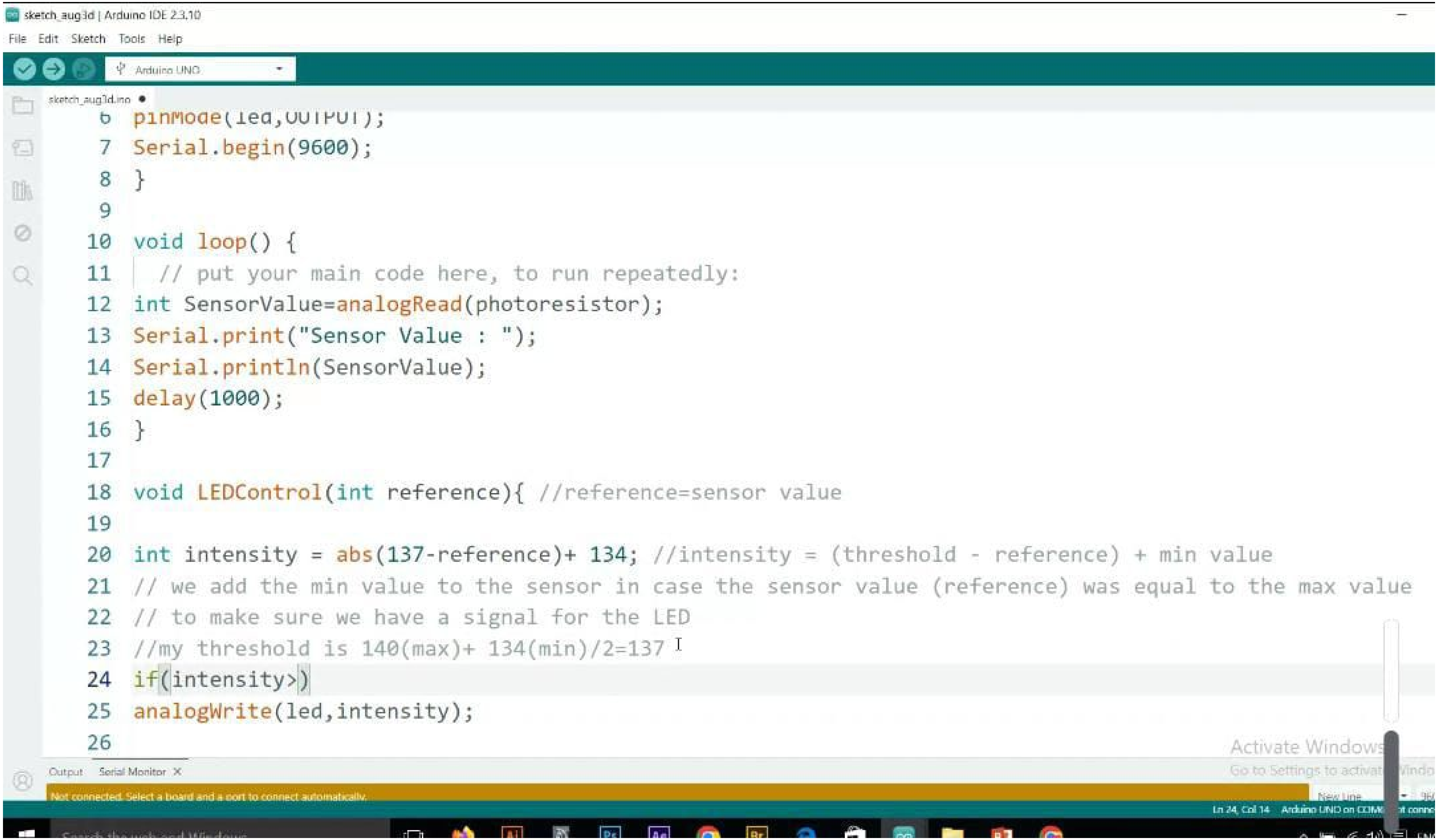
```
pinMode(led,OUTPUT);
Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
int SensorValue=analogRead(photoresistor);
Serial.print("Sensor Value : ");
Serial.println(SensorValue);
delay(1000);
}

void LEDControl(int reference){ //reference=sensor value

int intensity = abs(137-reference)+ 134; //intensity = (threshold r
// we add the min value to the sensor in case the sensor value (refer
// to make sure we have a signal for the LED
min)/2=137
```

Mohamad is presenting


sketch_aug3d | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_aug3d.ino
pinMode(led,OUTPUT);
Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
int SensorValue=analogRead(photoresistor);
Serial.print("Sensor Value : ");
Serial.println(SensorValue);
delay(1000);
}

void LEDControl(int reference){ //reference=sensor value

int intensity = abs(137-reference)+ 134; //intensity = (threshold - reference) + min value
// we add the min value to the sensor in case the sensor value (reference) was equal to the max value
// to make sure we have a signal for the LED
//my threshold is 140(max)+ 134(min)/2=137
if(intensity>)
analogWrite(led,intensity);
Output
Serial Monitor
Not connected. Select a board and a port to connect automatically.
Ln 24, Col 14
Activate Windows